

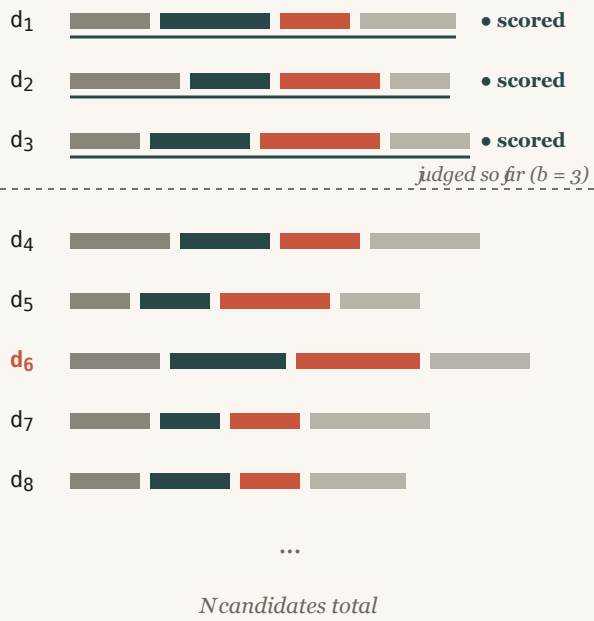
# ORE — Online Relevance Estimation

*A linear bandit learns which signals matter — per query, online*

## CANDIDATE POOL

*each doc carries a feature vector*

■ BM25 ■ dense ■ affinity ■ neighbor



*only a few will be sent to the expensive ranker*

## LINEAR BANDIT

$$\text{score}(d) = \alpha^T \phi(d)$$

### PARAMETERS $\alpha$

|                         |   |      |
|-------------------------|---|------|
| $\alpha_{\text{BM25}}$  | ■ |      |
| $\alpha_{\text{dense}}$ | ■ |      |
| $\alpha_{\text{aff}}$   | ■ | 0.31 |
| $\alpha_{\text{nbr}}$   | ■ | 0.05 |

— forward — selection

- - feedback — observed scores update  $\alpha$ , re-prioritize pool

## EXPENSIVE RANKER

*cross-encoder /  
RankLLaMA / etc.*

**ground-truth signal**